CHAPTER 5

Percentage

INTRODUCTION

The term *percent* means per hundred or for every hundred. It is the abbreviation of the Latin phrase *percentum*.

Scoring 60 per cent marks means out of every 100 marks the candidate scored 60 marks.

The term percent is sometimes abbreviated as p.c. The symbol % is often used for the term percent.

Thus, 40 percent will be written as 40%.

A fraction whose denominator is 100 is called a *percentage* and the numerator of the fraction is called *rate percent*, e.g., $\frac{5}{100}$ and 5 percent means the same thing, i.e., 5 parts out of every hundred parts.

BASIC FORMULAE

01 To convert a fraction into a percent:

To convert any fraction $\frac{l}{m}$ to rate percent, multiply it by 100 and put % sign, i.e., $\frac{l}{m} \times 100\%$

Illustration 1: What percentage is equivalent to $\frac{3}{5}$?

Solution: $\frac{3}{5} \times 100 = 60\%$.

02 To convert a per cent into a fraction:

To convert a percent into a fraction, drop the percent sign and divide the number by 100.

Illustration 2: What fraction is $16\frac{2}{3}\%$?

Solution: $16\frac{2}{3}\% = \frac{\left(\frac{50}{3}\right)}{100} = \left(\frac{50}{3} \times \frac{1}{100}\right) = \frac{1}{6}$.

03 To find a percentage of a given number:

$x\%$ of given number (N) $= \frac{x}{100} \times$ N.

Illustration 3: 75% of 400 = ?

Solution: 75% of 400 $= \frac{75}{100} \times 400 = 300$.

Illustration 4: Find a number whose 4% is 72.

Solution: Let the required number be x.

Then, 4% of $x = 72$

$\Rightarrow \quad \frac{4}{100} \times x = 72 \Rightarrow x = \frac{100}{4} \times 72 = 1800$.

Illustration 5: What per cent of 25 Kg is 3.5 Kg?

Solution: Let x% of 25 Kg be 3.5 Kg.

Then, x% of 25 Kg = 3.5 Kg

$\Rightarrow \quad \frac{x}{100} \times 25 = 3{\cdot}5 \Rightarrow x = \frac{3{\cdot}5 \times 100}{25} = 14$.

Hence, 3.5 Kg is 14% of 25 Kg.

SHORT-CUT METHODS

01 (a) If A is x% more than that of B, then B is less than that of A by

$$\left[\frac{x}{100+x} \times 100\right]\%$$

(b) If A is x% less than that of B, then B is more than that of A by

$$\left[\frac{x}{100-x} \times 100\right]\%$$

Explanation

Given: $A = B + \frac{x}{100}B = \frac{100+x}{100}B$

$\therefore \quad A - B = \frac{100+x}{100}B - B$

$= \left(\frac{100+x}{100} - 1\right)B = \frac{x}{100}B.$

So, $\frac{A-B}{A} = \frac{\frac{x}{100}B}{\frac{100+x}{100}B} = \frac{x}{100+x}$

$\Rightarrow \quad A - B = \left(\frac{x}{100+x} \times 100\right)\%$ of A.

Therefore, B is less than that of A by

$$\left(\frac{x}{100+x} \times 100\right)\%$$

Similarly, (b) can be proved.

Illustration 6: If Mohan's salary is 10% more than that of Sohan, then how much per cent is Sohan's salary less than that of Mohan?

Solution: Here, $x = 10$.

$\therefore$ Required answer $= \left(\frac{x}{100+x} \times 100\right)\%$

$= \left(\frac{x}{100+x} \times 100\right)\%$

$= 9\frac{1}{11}\%$

Illustration 7: If A's income is 40% less than B's income, then how much per cent is B's income more than A's income?

Solution: Here, $x = 40$.

$\therefore$ Required answer $= \left(\frac{x}{100-x} \times 100\right)\%$

$= \left(\frac{40}{100-40} \times 100\right)\%$

$= 66\frac{2}{3}\%$

02 If A is x% of C and B is y% of C, then

$$A = \frac{x}{y} \times 100\% \text{ of } B.$$

Explanation

Given: $A = \frac{x}{100}C \Rightarrow C = 100\frac{A}{x}$

and, $B = \frac{y}{100}C \Rightarrow C = 100\frac{B}{y}$

$\therefore \quad C = 100\frac{A}{x} = 100\frac{B}{y} \Rightarrow A = \frac{x}{y}B$

or, $\frac{x}{y} \times 100\%$ of B.

Illustration 8: If A is 20% of C and B is 25% of C, then what percentage is A of B?

Solution: Here, $x = 20$ and $y = 25$.

$A = \frac{x}{y} \times 100\%$ of B

$= \frac{20}{25} \times 100\%$ of B, i.e., 80% of B.

03 (a) If two numbers are respectively x% and y% more than a third number, then the first number is $\left(\frac{100+x}{100+y} \times 100\right)\%$ of the second and the second number is $\left(\frac{100+y}{100+x} \times 100\right)$ % of the first.

(b) If two numbers are, respectively, x% and y% less than a third number, then the first number is $\left(\frac{100-x}{100-y} \times 100\right)\%$ of the second and the second number is $\left(\frac{100-y}{100-x} \times 100\right)\%$ of the first number.

Explanation

Let A, B and C be the three numbers.

Given:

$$A = C + \frac{x}{100}C = \left(\frac{100+x}{100}\right)C \Rightarrow C = A\left(\frac{100}{100+x}\right)$$

and, $B = C + \frac{y}{100}C = \left(\frac{100+y}{100}\right)C \Rightarrow C = B\left(\frac{100}{100+y}\right)$

$\therefore \quad A\left(\frac{100}{100+x}\right) = B\left(\frac{100}{100+y}\right)$

$\Rightarrow \quad A = \left(\frac{100+x}{100+y}\right)B$ or $\left(\frac{100+x}{100+y}\right) \times 100\%$ of B

and, $B = \left(\frac{100+y}{100+x}\right)A$ or $\left(\frac{100+y}{100+x}\right)$ × 100% of A.

Similarly, (b) can be proved.

Illustration 9: Two numbers are respectively 20% and 50% more than a third number. What per cent is the first of the second?

Solution: Here, $x = 20$ and $y = 50$.

$$\therefore \quad \text{First number} = \left(\frac{100+x}{100+y}\right) \times 100\% \text{ of the second}$$

$$= \left(\frac{100+20}{100+50}\right) \times 100\% \text{ of the second}$$

i.e., 80% of the second.

Illustration 10: Two numbers are, respectively, 32% and 20% less than a third number. What per cent is the first of the second?

Solution: Here, $x = 32$ and $y = 20$.

$$\therefore \quad \text{First number} = \left(\frac{100-x}{100-y}\times 100\right)\% \text{ of the second}$$

$$= \left(\frac{100-32}{100-20}\times 100\right)\% \text{ of the second}$$

i.e., 85% of the second.

04 (a) If the price of a commodity increases by P%, then the reduction in consumption so as not to increase the expenditure is

$$\left(\frac{P}{100+P}\times 100\right)\%$$

(b) If the price of a commodity decreases by P%, then the increase in consumption so as not to decrease the expenditure is

$$\left(\frac{P}{100-P}\times 100\right)\%$$

Explanation

Let the original price of the commodity be ₹100.

Then, the increased price = $100 + \frac{P}{100}\times 100$

= ₹$(100 + P)$.

Therefore, to keep the price unchanged, there should be a reduction in the consumption of the commodity by ₹P.

$\because$ Decrease in ₹$(100 + P)$ = ₹P

$\therefore$ Decrease in ₹100 = $\frac{P}{100+P}\times 100$

$\therefore$ Required reduction in consumption is

$$\left(\frac{P}{100+P}\times 100\right)\%$$

Similarly, (b) part can be proved.

Illustration 11: If the price of sugar increases by 25%, find how much per cent its consumption be reduced so as not to increase the expenditure.

Solution: Reduction in consumption

$$= \left(\frac{P}{100+P}\times 100\right)\%$$

$$= \left(\frac{25}{100+25}\times 100\right)\% \text{ or } 20\%$$

Illustration 12: If the price of a commodity decreases by 25%, find how much per cent its consumption be increased so as not to decrease the expenditure.

Solution: Increase in consumption

$$= \left(\frac{P}{100-P}\times 100\right)\%$$

$$= \left(\frac{25}{100-25}\times 100\right)\% \text{ or } 33\frac{1}{3}\%$$

05 If a number is changed (increased/decreased) successively by x% and y%, then net % change is given by $\left(x+y+\frac{xy}{100}\right)$% which represents increase or decrease in value according as the sign is +ve or –ve.

If x or y indicates decrease in percentage, then put –ve sign before x or y, otherwise +ve sign.

Explanation

Let the given number be N.

If it is increased by x%, then it becomes

$$N + x\% \text{ of } N = N + \frac{Nx}{100} = \frac{N(x+100)}{100}.$$

If it is further increased by y%, then it becomes

$$\frac{N(x+100)}{100} + \frac{y}{100}\times\frac{N(x+100)}{100}$$

$$= \frac{N(x+100)(y+100)}{(100)^2}$$

$$\therefore \quad \text{Net change} = \frac{N(x+100)(y+100)}{(100)^2} - N$$

$$= \frac{N(100x+100y+xy)}{(100)^2}$$

$$\therefore \quad \%\text{ change} = N\left(x+y+\frac{xy}{100}\right)\times\frac{1}{100}\times\frac{100}{N}$$

$$= \left(x + y + \frac{xy}{100}\right)\%$$

Illustration 13: If salary of a person is first increased by 15% and thereafter decreased by 12%, what is the net change in his salary?

Solution: Here, $x = 15$ and $y = -12$.

$\therefore$ The net% change in the salary

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(15 - 12 - \frac{15 \times 12}{100}\right)\% \text{ or } 1.2\%$$

Since the sign is +ve, the salary of the person increases by 1.2%

Illustration 14: The population of a town is decreased by 25% and 40% in two successive years. What per cent population is decreased after two years?

Solution: Here, $x = -25$ and $y = -40$.

$\therefore$ The net% change in population

$$= \left(x + y + \frac{xy}{100}\right)\%$$

$$= \left(-25 - 40 + \frac{25 \times 40}{100}\right)\% \text{ or } -55\%$$

Since the sign is –ve, there is decrease in population after two years by 55%

06 If two parameters A and B are multiplied to get a product and if A is changed (increased/decreased) by x% and another parameter B is changed (increased/decreased) by y%, then the net % change in the product ($A \times B$) is given $\left(x + y + \frac{xy}{100}\right)\%$ which represents increase or decrease in value according as the sign in +ve or –ve.

If x or y indicates decrease in percentage, then put –ve sign before x or y, otherwise +ve sign.

Illustration 15: If the side of a square is increased by 20%, its area is increased by k%. Find the value of k.

Solution: Since side × side = area

$\therefore$ Net% change in area

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(20 + 20 + \frac{20 \times 20}{100}\right)\%$$

[Here, $x = 20$ and $y = 20$]

$= 44\%$

Therefore, the area is increased by 44%.

Here, $k = 44$.

Illustration 16: The radius of a circle is increased by 2%. Find the percentage increase in its area.

Solution: Since $\pi \times$ radius × radius = area

$\therefore$ Net% change in area

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(2 + 2 + \frac{2 \times 2}{100}\right)\%$$

[Here, $x = 2$ and $y = 2$]

$$= 4\frac{1}{25}\%$$

Therefore, the percentage increase in area is $4\frac{1}{25}\%$

Illustration 17: The tax on a commodity is diminished by 15% and its consumption increases by 10%. Find the effect on revenue.

Solution: Since tax × consumption = revenue

$\therefore$ Net% change in revenue

$$= \left(x + y + \frac{xy}{100}\right)\% = \left(-15 + 10 - \frac{15 \times 10}{100}\right)\%$$

[Here, $x = -15$ and $y = 10$]

$= -6.5\%$

$\therefore$ The revenue decreases by 6.5%

07 If the present population of a town (or value of an item) be P and the population (or value of item) changes at r% per annum, then

(a) Population (or value of item) after n years

$$= P\left(1 + \frac{r}{100}\right)^n$$

(b) Population (or value of item) n years ago

$$= \frac{P}{\left(1 + \frac{r}{100}\right)^n}.$$

where r is +ve or –ve according as the population (or value of item) increases or decreases.

Explanation

Population at the end of first year

$$= P + \frac{r}{100}P = P\left(1 + \frac{r}{100}\right).$$

Now, the population at the beginning of second year

$$= P\left(1 + \frac{r}{100}\right).$$

$\therefore$ Population at the end of second year

$$= P\left(1+\frac{r}{100}\right)+\frac{r}{100}P\left(1+\frac{r}{100}\right) = P\left(1+\frac{r}{100}\right)^2$$

$\vdots \qquad \vdots \qquad \vdots$

Population at the end of n years $= P\left(1+\frac{r}{100}\right)^n$.

Illustration 18: The population of a town increases 5% annually. If its present population is 84000, what will it be in 2 years time?

Solution: Here, $P = 84000$, $r = 5$ and $n = 2$.

$\therefore$ Population of the town after 2 years

$\because$

$$= P\left(1+\frac{r}{100}\right)^n = 84000\left(1+\frac{5}{100}\right)^2$$

$$= 84000 \times \frac{105}{100}\times\frac{105}{100} = 92610.$$

Illustration 19: The population of a town increases at the rate of 5% annually. If the present population is 4410, what it was 2 years ago?

Solution: Here, $P = 4410$, $r = 5$ and $n = 2$.

$\therefore$ Population of the town 2 years ago

$$= \frac{P}{\left(1+\frac{r}{100}\right)^n} = \frac{4410}{\left(1+\frac{5}{100}\right)^2} = \frac{4410}{\frac{105}{100}\times\frac{105}{100}} = 4000.$$

08 If a number A is increased successively by x% followed by y% and then by z%, then the final value of A will be

$$A\left(1+\frac{x}{100}\right)\left(1+\frac{y}{100}\right)\left(1+\frac{z}{100}\right)$$

In case a given value decreases by any percentage, we will use a –ve sign before that.

Illustration 20: The population of a town is 144000. It increases by 5% during the first year. During the second year, it decreases by 10% and increases by 15% during the third year. What is the population after 3 years?

Solution: Here, $P = 144000$, $x = 5$, $y = -10$ and $z = 15$.

$\therefore$ Population of the town after 3 years

$$= A\left(1+\frac{x}{100}\right)\left(1+\frac{y}{100}\right)\left(1+\frac{z}{100}\right)$$

$$= 144000\left(1+\frac{5}{100}\right)\left(1-\frac{10}{100}\right)\left(1+\frac{15}{100}\right)$$

$$= \frac{144000\times105\times90\times115}{100\times100\times100} = 156492.$$

09 In an examination, the minimum pass percentage is x% If a student secures y marks and fails by z marks, then the maximum marks in the examination is $\frac{100(y+z)}{x}$.

Explanation

Let the maximum marks be m.

Given: x% of $m = y + z$

$$\Rightarrow \quad \frac{x}{100} \times m = y + z \text{ or } m = \frac{100(y+z)}{x}.$$

Illustration 21: In an examination, a student must get 60% marks to pass. If a student who gets 120 marks, fails by 60 marks, find the maximum marks.

Solution: Here, $x = 60$, $y = 120$ and $z = 60$.

$\therefore$ Maximum marks

$$= \frac{100(y+z)}{x} = \frac{100(120+60)}{60} = \frac{100\times180}{60} = 300.$$

10 In an examination x% and y% students respectively fail in two different subjects while z% students fail in both the subjects, then the percentage of students who pass in both the subjects will be $(100 - (x + y - z))$%

Explanation

Percentage of students who failed in one subject = $(x - z)$%

Percentage of students who failed in other subject = $(y - z)$%

Percentage of students who failed in both the subjects = z%

$\therefore$ Percentage of students who passed in both the subjects

$$= [100 - [(x - z) + (y - z) + z]]\%$$
$$= (100 - (x + y - z))\%$$

Illustration 22: In an examination, 42% students failed in Mathematics and 52% failed in Science. If 17% failed in both the subjects, find the percentage of those who passed in both the subjects.

Solution: Here, $x = 42$, $y = 52$ and $z = 17$.

$\therefore$ Percentage of students passing both the subjects

$$= (100 - (x + y - z))\%$$
$$= (100 - (42 + 52 - 17))\% \text{ or } 23\%$$

EXERCISE-I

1. What percentage is equivalent to $5\frac{1}{4}$?

(a) 525% (b) 425%
(c) 625% (d) None of these

2. 0.005 = (... ? ...)%

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$%
(c) $\frac{1}{3}$% (d) None of these

3. $6\frac{2}{3}$% expressed as a fraction in its lowest term is:

(a) $\frac{2}{15}$ (b) $\frac{1}{15}$
(c) $\frac{3}{20}$ (d) None of these

4. What fraction is 0.6%

(a) $\frac{7}{500}$ (b) $\frac{9}{500}$
(c) $\frac{3}{500}$ (d) None of these

5. 0.025 in terms of rate per cent is:

(a) 3.5% (b) 2.5%
(c) 1.5% (d) None of these

6. What per cent of 12 is 84?

(a) 800% (b) 600%
(c) 700% (d) None of these

7. Express $\frac{7}{8}$ as percentage.

(a) $67\frac{1}{2}$ (b) $87\frac{1}{2}$%
(c) $97\frac{1}{4}$% (d) None of these

8. Express $8\frac{1}{3}$% as a fraction.

(a) $\frac{1}{12}$ (b) $\frac{1}{16}$
(c) $\frac{1}{18}$ (d) None of these

9. $37\frac{1}{2}$% of ₹48 is:

(a) ₹20 (b) ₹16
(c) ₹18 (d) None of these

10. What per cent of $\frac{2}{7}$ is $\frac{1}{35}$?

(a) 15% (b) 18%
(c) 10% (d) None of these

11. 75% of 480 = (?) × 15

(a) 12 (b) 36
(c) 24 (d) None of these

12. If 200% of a number is 90, then what is the 80% of that number?

(a) 48
(b) 36
(c) 24
(d) None of these

13. If $37\frac{1}{2}$% of a number is 45, then $87\frac{1}{2}$% of the number will be:

(a) 115 (b) 135
(c) 105 (d) None of these

14. ? × 15 = 37.5% of 220.

(a) 5.5 (b) 7.5
(c) 6.5 (d) None of these

15. What per cent of 4 Km is 8 metres?

(a) 0.4 (b) 0.2
(c) 0.8 (d) None of these

16. x% of y + y% of x = ?

(a) 3% of xy
(b) 2% of xy
(c) 5% of xy
(d) None of these

17. 0.35% of a number is equivalent to multiplying it by the number:

(a) 0.0025 (b) 0.0045
(c) 0.0035 (d) None of these

18. If 8% of x = 4% of y, then 20% of x is:

(a) 15% of y (b) 10% of y
(c) 20% of y (d) None of these

19. x% of y + ?% of x = x% of (x + y).

(a) $x + y$ (b) x
(c) y (d) None of these

20. A number x is 125% of y. To compute y, the number x has to be multiplied by:

(a) 0.08 (b) 0.4
(c) 0.8 (d) None of these

21. 25% of 25% = ?

(a) 6.25 (b) 0.0625
(c) 0.625 (d) None of these

22. Which number is 60% less than 80?

(a) 24 (b) 36
(c) 32 (d) None of these

23. 20% of 30% of 20% of ₹850 is:

(a) ₹9.50 (b) ₹10.20
(c) ₹10.50 (d) None of these

24. The greatest of $16\frac{2}{3}$%, $6\frac{2}{3}$%, 0·3 is:

(a) $16\frac{2}{3}$% (b) $6\frac{2}{3}$%
(c) 0.3 (d) Cannot be compared

25. 40% of 20% + 30% of 25% + 50% of 28% is equivalent to:

(a) 29.5% (b) 28.5%
(c) 30.5% (d) None of these

26. If 90% of A = 30% of B and B = x% of A, then the value of x is:

(a) 800 (b) 300
(c) 700 (d) None of these

27. 1 quintal 25 Kg is what per cent of 1 metric tonne?

(a) $16\frac{1}{2}$% (b) $8\frac{1}{2}$%
(c) $12\frac{1}{2}$% (d) None of these

28. If 12% of x is equal to 6% of y, then 18% of x will be equal to how much per cent of y?

(a) 7% (b) 9%
(c) 11% (d) None of these

29. If a number is 20% more than the other, how much per cent is the second number less than the first?

(a) $12\frac{1}{3}$% (b) $16\frac{2}{3}$%
(c) $16\frac{1}{3}$% (d) None of these

30. If A's income is 25% less than that of B, then how much per cent is B's income more than that of A?

(a) $33\frac{1}{3}$% (b) $66\frac{2}{3}$%
(c) $11\frac{2}{3}$% (d) None of these

31. If the given two numbers are respectively 7% and 28% of a third number, then what percentage is the first of the second?

(a) 20% (b) 25%
(c) 18% (d) None of these

32. Two numbers are respectively 60% and 20% more than a third number. Second number expressed as a percentage of first is:

(a) 75% (b) 90%
(c) 80% (d) None of these

33. Two numbers are less than a third number by 30% and 37%, respectively. How much per cent is the second number less than the first?

(a) 15%
(b) 10%
(c) 20%
(d) None of these

34. Two numbers are respectively 20% and 10% more than a third number. How much per cent is the first number more than the second?

(a) $9\frac{1}{11}$% (b) $7\frac{1}{11}$%
(c) $11\frac{1}{11}$% (d) None of these

35. The price of cooking oil has increased by 15%. The percentage of reduction that a family should effect in the use of cooking oil so as not to increase the expenditure on this account is:

(a) $15\frac{2}{23}$% (b) $13\frac{1}{23}$%
(c) $17\frac{1}{23}$% (d) None of these

36. If the price of apples goes down by 10%, find the percentage of increase that a family should effect in its consumption so as not to increase expenditure on this account is:

(a) $13\frac{1}{11}$% (b) $15\frac{1}{11}$
(c) $11\frac{1}{11}$% (d) None of these

37. A number is increased by 20% and then decreased by 20%, the final value of the number:

(a) Does not change
(b) Decreases by 2%
(c) Increases by 4%
(d) Decreases by 4%.

38. A man's wages were decreased by 50%. Again, the reduced wages were increased by 50%. He has a loss of:

(a) 35% (b) 25%
(c) 20% (d) None of these

39. The population of a town is decreased by 20% and 25% in two successive years. What per cent population is decreased after two years?

(a) 50% (b) 40%
(c) 60% None of these

40. The difference between a discount of 35% and two successive discounts of 20% and 20% on a certain bill was ₹22. Find the amount of the bill.

(a) ₹3200 (b) ₹2200
(c) ₹1800 (d) None of these

41. A shopkeeper marks the prices of his goods at 25% higher than the original price. After that, he allows a discount of 12%. What profit or loss did he make?

(a) 10% profit (b) 15% profit
(c) 10% loss (d) 15% loss

42. Two shopkeepers sell a ratio of similar brand and type at the same list price of ₹1000. The first allows two successive discounts of 20% and 10% and the second allows the successive discounts of 15% and 15%. Find the difference in discounts offered by the two shopkeepers.

(a) ₹3.50 (b) ₹1.50
(c) ₹2.50 (d) None of these

43. The tax on a commodity is diminished by 10% and its consumption increases by 10%. Find the effects on revenue.

(a) 1% (b) 2%
(c) 3% (d) None of these

44. The radius of a sphere is increased by 10%. The surface area increases by

(a) 21% (b) 31%
(c) 41% (d) None of these

45. When the price of an article is reduced by 15%, the sales increases by 35%. The percentage change in the total amount of receipts is:

(a) $14\frac{3}{4}$% decrease (b) $14\frac{3}{4}$% increase
(c) $13\frac{3}{4}$% decrease (d) None of these

46. If the side of a square is increased by 30%, its area is increased by:

(a) 49% (b) 69%
(c) 79% (d) None of these

47. The length and breadth of a square are increased by 30% and 20%, respectively. The area of the rectangle so formed exceeds the area of the square by:

(a) 56% (b) 46%
(c) 66% (d) None of these

48. In measuring the sides of a rectangle, one side is taken 10% in excess and the other 20% in deficit. Find the error per cent in area calculated from the measurement.

(a) 12% deficit (b) 10% deficit
(c) 12% excess (d) None of these

49. For a rectangle, the length and breadth are increased by 10% and 20%, respectively. The percentage increase in area is:

(a) 24% (b) 48%
(c) 32% (d) None of these

50. Water tax is increased by 20% but its consumption is decreased by 20%. The increase or decrease in the expenditure is:

(a) 4% decrease
(b) 4% increase
(c) 8% decrease
(d) 8% increase

51. On decreasing the price of a colour TV by 30%, its sale is increased by 20%. The effect on the revenue is:

(a) 16% decrease (b) 16% increase
(c) 20% increase (d) None of these

52. The population of a city increases at the rate of 10% annually. Its present population is 90.51 lacs. The population 3 years ago was nearly:

(a) 72 Lakhs (b) 68 Lakhs
(c) 80 Lakhs (d) None of these

53. The value of a machine depreciates at the rate of 10% every year. It was purchased 3 years ago. If its present value is ₹8748, its purchase price was:

(a) ₹16000 (b) ₹18000
(c) ₹12000 (d) None of these

54. The income of a company increases 20% per annum. If its income is ₹2664000 in the year 1999 what was its income in the year 1997?

(a) ₹1750000 (b) ₹1650000
(c) ₹1850000 (d) None of these

55. The population of a town is 32000. It increases 15% annually. What will it be in 2 years?

(a) 52340 (b) 42320
(c) 62430 (d) None of these

56. The value of a machine is ₹6250. It decreases by 10% during the first year, 20% during the second year and 30% during the third year. What will be the value of the machine after 3 years?

(a) ₹2650 (b) ₹3050
(c) ₹3150 (d) None of these

57. The population of a town increases by 12% during first year and decreases by 10% during second year. If the present population is 50400, what it was 2 years ago?

(a) 40000 (b) 35000
(c) 50000 (d) None of these

58. Ramesh loses 20% of his pocket money. After spending 25% of the remainder he has ₹480 left. What was his pocket money?

(a) ₹600 (b) ₹800
(c) ₹900 (d) None of these

59. An army lost 10% its men in war, 10% of the remaining due to diseases and 10% of the rest were hurt. Thus, the strength was reduced to 729000 active men. Find the original strength.

(a) 1000000 (b) 1200000
(c) 1500000 (d) None of these

60. The daily wage is increased by 25% and a person now gets ₹25 per day. What was his daily wage before the increase?

(a) ₹25 (b) ₹20
(c) ₹30 (d) None of these

61. A student has to secure 15% marks to get through. If he gets 80 marks and fails by 70 marks, find the maximum marks set for the examination.

(a) 900 (b) 1000
(c) 1200 (d) None of these

62. In an examination, 30% and 35% students respectively failed in History and Geography while 27% students failed in both the subjects. If the number of students passing the examination is 248, find the total number of students who appeared in the examination.

(a) 425 (b) 380
(c) 400 (d) None of these

63. Mr Katial buys a house for ₹100000 and rents it. He puts 12.5% of each month's rent aside for upkeep and repairs, pays ₹325 per year as taxes and realizes 5.5% annually on his investment. Find the monthly rent.

(a) ₹634.76 (b) ₹554.76
(c) ₹654.76 (d) None of these

64. In an examination, there were 2000 candidates, out of which 900 candidates were boys and rest were girls. If 32% of the boys and 38% of the girls passed, then the total percentage of failed candidates is:

(a) 35.3% (b) 64.7%
(c) 68.5% (d) 70%

65. From the salary of an officer, 10% is deducted as house rent, 15% of the rest he spends on children's education and 10% of the balance, he spends on clothes. After this expenditure he is left with ₹1377. His salary is:

(a) ₹2000 (b) ₹2040
(c) ₹2100 (d) ₹2200

66. If the price of gold increases by 30%, find by how much the quantity of ornaments must be reduced so that the expenditure may remain the same as before?

(a) $27\frac{2}{13}\%$ (b) $23\frac{1}{13}\%$
(c) 30% (d) 19%

67. The price of sugar has fallen by 10%. How many quintals can be bought for the same money which was sufficient to buy 18 quintals at the higher price?

(a) 20 (b) 22
(c) 25 (d) 30

68. In an examination, there are 1000 boys and 800 girls, 60% of boys and 40% girls passed. The percentage of candidates that failed is:

(a) 48.88 (b) 45.88
(c) 50.00 (d) 49.88

69. The price of an article is cut by 20%. To restore it to its original price, the new price must be increased by:

(a) 20% (b) $22\frac{1}{2}\%$
(c) 25% (d) 40%

70. In a fraction, numerator is increased by 25% and the denominator is diminished by 10%. The new fraction obtained is $\frac{5}{9}$. The original fraction is:

(a) $\frac{2}{5}$ (b) $\frac{5}{9}$

(c) $\frac{3}{5}$ (d) None of these

71. One side of a square is increased by 30%. To maintain the same area the other side will have to be decreased by:

(a) $23\frac{1}{13}\%$ (b) $76\frac{12}{13}\%$

(c) 30% (d) 15%

EXERCISE-2
(BASED ON MEMORY)

1. 0.06% of 250% of 1600 is

(a) 24 (b) 0.24

(c) 0.024 (d) 2.4

[SSC CGL Tier-II CBE, 2018]

2. Two numbers are 90% and 75% lesser than a third number. By what % should the first number be increased so that it becomes equal to the second number?

(a) 250 (b) 200

(c) 150 (d) 100

[SSC CGL Tier-II CBE, 2018]

3. When a number is increased by 216, it becomes 140% of itself. What is the number?

(a) 540 (b) 756

(c) 450 (d) 875

[SSC CGL Tier-II CBE, 2018]

4. If $X = 600$ and $Y = 800$, then X is how much per cent less than Y?

(a) 33.33 (b) 25

(c) 75 (d) 35

[SSC CHSL (10 + 2) Tier-I CBE, 2018]

5. A man donates 30% of his wealth to charity. 30% and 25% of the remaining wealth to his wife and son respectively. The rest he divides equally between his three daughters. One of his daughter gets ₹ 42 lakh as her share. What was the man's wealth the (₹ lakhs)?

(a) 280 (b) 400

(c) 500 (d) 350

[SSC CGL Tier-II CBE, 2018]

6. A man's annual income has increased by ₹ 5 lakhs but the tax on income that he has to pay has reduced from 12% to 10%. He now pays ₹ 10,000 more income tax. What is his increased income (in ₹ lakhs)

(a) 20 (b) 25

(c) 15 (d) 10

[SSC CGL Tier-II CBE, 2018]

7. 40% are the passing marks. A students gets 250 marks yet fails by 38 marks. What is the maximum marks?

(a) 720 (b) 750

(c) 800 (d) 840

[SSC CGL Tier-II CBE, 2018]

8. Ravi is 12 years younger than Surya. Ravi's age 40% of the sum of his and Surya's age. What will be Surya's age 9 years hence?

(a) 36 (b) 24

(c) 33 (d) 45

[SSC CGL Tier-II CBE, 2018]

9. A boy found the answer for the question "subtract the sum of $\frac{1}{4}$ and $\frac{1}{5}$ from unity and express the answer in decimals" as 0.45. The percentage of error in his answer was

(a) $\left(\frac{100}{11}\right)\%$ (b) 50%

(c) 10% (d) $\left(\frac{200}{11}\right)\%$

[SSC CGL Tier-II CBE, 2017]

10. $a\%$ of $b + b\%$ of $a =$ _____

(a) $2a\%$ of b (b) $2a\%$ of $2b$

(c) $2a\%$ of $2a$ (d) $2b\%$ of $2b$

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

11. A number is increased by 84, it becomes 107% of itself. What is the number?

(a) 600 (b) 900

(c) 1500 (d) 1200

[SSC CGL Tier-I (CBE), 2017]

12. 20% of $a = b$, then b% of 20 is the same as _____ of a

(a) 8% (b) 40%
(c) 4% (d) 80%

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

13. A man wills 40% of his wealth to his wife and rest to his children. What per cent of the wealth willed to the wife do the children get?

(a) 150 (b) 66.6
(c) 50 (d) 20

[SSC CGL Tier-I CBE, 2017]

14. When 40 is subtracted from 30% of a number, the result is 50. What is the value of the numberr?

(a) 150 (b) 180
(c) 300 (d) 450

[SSC Multi-Tasking Staff, 2017]

15. If 18% of a number is 720, then what is 81% of the same number?

(a) 2280 (b) 1620
(c) 3240 (d) 3160

[SSC Multi-Tasking Staff, 2017]

16. Two numbers are 10% and 25% lesser than a third number. By how much percent is the second number to be enhanced to make it equal to the first number?

(a) 16.67 per cent
(b) 20 per cent
(c) 15 per cent
(d) 60 per cent

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

17. P is 20% more than Q and 40% less than R. If value of Q is ₹150, then what is the value of R (in ₹)?

(a) 300 (b) 320
(c) 220 (d) 250

[SSC CAPFs ASI & Delhi Police SI, 2017]

18. If 25% of a number is 6, what is the number is 50% more than the initial number?

(a) 36 (b) 24
(c) 30 (d) 16

[SSC CAPFs ASI & Delhi Police SI, 2017]

19. A, B, C are three students. A got 18% more marks than B and 12% less than C. If B got 220 marks, how much marks C has got?

(a) 230 (b) 295
(c) 240 (d) 290

[SSC CGL Tier-I CBE, 2017]

20. If A is 20% more than 360, then what is the value of A?

(a) 396 (b) 432
(c) 480 (d) 512

[SSC Multi-Tasking Staff, 2017]

21. x is 30% more than y and 25% less than z. If value of y is ₹ 300, then what is the value (in ₹) of z?

(a) 390 (b) 400
(c) 470 (d) 520

[SSC CAPFs ASI & Delhi Police SI, 2017]

22. Two numbers are more than the third number by 20% and 80% respectively. First number is what per cent of the second number?

(a) 75 (b) 50
(c) 66.66 (d) 150

[SSC Delhi Police Constable, 2017]

23. If income of R is 30% more than the income of Q and the income of Q is 20% more than the income of P the income of R is how much percent more than the income of P?

(a) 50 (b) 10
(c) 56 (d) 55

[SSC Multi-Tasking Staff, 2017]

24. The average monthly salary of all the employees in a factory is ₹ 8840. If the average salary of all the officers is ₹ 15000 and that of the remaining employees is ₹ 8000, then what is the percentage of the officers among the employees?

(a) 10% (b) 12%
(c) $8\frac{1}{3}\%$ (d) 11%

[SSC CGL Tier-II CBE, 2017]

25. The monthly salary of Mr. Sachdev gets increased by 5% there by his salary becomes ₹ 15,120 per annum. His earlier monthly salary (before the increase) was

(a) ₹ 1,320 (b) ₹ 1,200
(c) ₹ 1,240 (d) ₹ 1,440

[SSC Multi-Tasking Staff, 2017]

26. A person spends 25% of his annual income on house rent, 15% on education of children and 45% on other items. If he saves ₹ 14,400 annually, then the person's total income is:

(a) ₹ 96,000 (b) ₹ 98,000
(c) ₹ 1,00,000 (d) ₹ 1,20,000

[SSC Matric Level MTS, 2017]

27. Amit donated 205 of his income to a school and deposited 20% of the remainder in his bank. If he has ₹ 12800 now, what is the income (in ₹) of Amit?

(a) 18000 (b) 20000
(c) 24000 (d) 32000

[SSC CGL Tier-I CBE, 2017]

28. An engineering student has to secure 25% marks to pass. He gets 47 and fails by 43 marks. What are the examination?

(a) 385 marks (b) 410 marks
(c) 360 marks (d) 435 marks

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

29. Two students appeared for an examination. One of them secured 19 marks more than the other and his marks were 60% of the sum of their marks. The marks obtained by them are

(a) 78 and 59 (b) 57 and 38
(c) 45 and 26 (d) 99 and 80

[SSC CGL Tier-I CBE, 2017]

30. A person scores 45% of the total marks in the exam and still fails by 40 marks. The passing percentage of the exam is 55%. What is the maximum marks of the exam?

(a) 300 (b) 350
(c) 400 (d) 500

[SSC CGL Tier-I CBE, 2017]

31. The price of electricity has been increased by 25%. If a person wants to keep the expenditure same, then the percentage reduction in use of electricity should be:

(a) 19 (b) 18
(c) 21 (d) 20

[SSC Matric Level, 2017]

32. If the price of apple increase from ₹ 80 per kg to ₹ 100 per kg, by what per cent a person should decrease the consumption of apple so that his expenditure remains same?

(a) 25 (b) 22
(c) 20 (d) 18

[SSC CAPFs ASI & Delhi Police SI, 2017]

33. The price of a house increase every year by 20%. If the present price is ₹ 720000, then what was its price (in ₹) 2 years ago

[SSC Multi-Tasking Staff, 2017]

34. The population of a city increases at the rate of 5% per annual. If the present population of the city is 3,70,440, its population years ago was:

(a) 30,000 (b) 2,80,000
(c) 3,60,000 (d) 3,20,000

[SSC Matric Level MTS, 2017]

35. the price of table depreciates every year by 20%. If the value of the table after 2 years will ₹ 32000, then what is present price (in ₹) of the table?

(a) 48000 (b) 44000
(c) 50000 (d) 51000

[SSC CAPFs ASI & Delhi Police SI, 2017]

36. The present population of a is 26010. It increase annual at the rate of 2%. What was the population of town two years ago?

(a) 25000 (b) 25100
(c) 25200 (d) 25500

[SSC CGL Tier-I CBE, 2017]

37. The price of motor cycle depreciates every year by 10%. If the value of the motor cycle after 3 years will be ₹ 36450, then what is the present value (in ₹) of the motor cycle?

(a) 45000 (b) 50000
(c) 48000 (d) 51000

[SSC CAPFs ASI & Delhi Police SI, 2017]

38. On a certain date, Pakistan has a success rate of 60% against India in all the ODIs played between the two countries. They lost the next 30 ODIs in a row to India and their success rate comes down to 30%. The total number of ODIs played between the two countries is

(a) 50 (b) 45
(c) 60 (d) 30

[SSC CGL Tier-II CBE, 2017]

39. A businessman imported Laptops, worth ₹ 210000, Mobile phones worth ₹ 100000 and Television sets worth ₹ 150000. He had to pay 10% duty on laptops, 8% on phones and 5% on Television sets as a special case. How much total duty (in Rupees) he had to pay on all items as per above details?

(a) 36500 (b) 37000
(c) 37250 (d) 37500

[SSC CGL Tier-II CBE, 2017]

40. The weight of two iron balls are 3.5 kg and 7.5 kg. What is the percentage weight of the first ball with respect to second ball?

(a) $46\frac{2}{3}\%$ (b) 35%

(c) $46\frac{1}{3}\%$ (d) 45%

[SSC CGL Tier-I CBE, 2016]

41. One number is 25% of another number. The larger number is 12 more than the smaller. The larger number is

(a) 48 (b) 16
(c) 4 (d) 12

[SSC CGL Tier-I (CBE), 2016]

42. The number of students in a class is increased by 20% and the number now becomes 66. Initially the number was

(a) 45 (b) 50
(c) 55 (d) 60

[SSC CGL Tier-I (CBE), 2016]

43. A village lost 12% of its goals in flood and 5% of remainder died from diseases. If the number left now is 8360, what was the original number before the flood?

(a) 1000 (b) 10000
(c) 1,00,000 (d) 8360

[SSC CGL Tier-II CBE, 2016]

44. If A is equal to 20% of B and B is equal to 25% of C; then what percent of C is equal to A?

(a) 10 (b) 15
(c) 5 (d) 20

[SSC CGL Tier-I (CBE), 2016]

45. In a school there are 1500 students, 44% of them are girls. Monthly fee of each boy is ₹ 540 and the fee of each girl is 25% less than that of a boy. The sum of fees of boys and girls both is

(a) ₹ 720600 (b) ₹ 720800
(c) ₹ 720900 (d) ₹ 721000

[SSC CGL Tier-I (CBE), 2016]

46. In a marriage party 32% are women 54% are men and there are 196 children. How many men are there in the marriage party?

(a) 756 (b) 448
(c) 332 (d) 324

[SSC CGL Tier-I (CBE), 2016]

47. $6\frac{1}{4}\%$ of 1600 + $12\frac{1}{2\%}$ of 800 equals

(a) 100 (b) 200
(c) 300 (d) 400

[SSC CGL Tier-I (CBE), 2016]

48. If the salary of Manoj is 40% less than that of subhash, then by how much percentage is the salary of Subhash more than that of Manoj?

(a) 60% (b) $66\frac{1}{4}\%$
(c) $66\frac{2}{3}\%$ (d) 65%

[SSC CGL Tier-I CBE, 2016]

49. Two numbers are 50% and 80% less than a third number. By how much per cent is the second number to be enhanced to make it equal to the first number?

(a) 150 per cent
(b) 60 per cent
(c) 30 per cent
(d) 37.5 per cent

[SSC CGL Tier-I CBE, 2016]

50. There is a ratio of 5 : 4 between two numbers. If 40% of the first number is 12, then what would be 50% of the second number?

(a) 12 (b) 24
(c) 18 (d) Data Inadequate

[SSC CPO SI, ASI, 2016]

51. If 10% of x is 3 times 15% of y, then find $x : y$.

(a) 7 : 2 (b) 9 : 2
(c) 8 : 3 (d) 11 : 4

[SSC CGL Tier-I CBE, 2016]

52. If 35% of A's income is equal to 25% of B's income, then the ratio of A's income to B's income is

(a) 7 : 5 (b) 5 : 7
(c) 4 : 7 (d) 4 : 3

[SSC CGL Tier-I CBE, 2016]

53. Mukesh has twice as much money as Soham. Soham has 50% more money than Pankaj. If the average money with them is ₹ 110, then Mukesh has

(a) ₹ 155 (b) ₹ 160
(c) ₹ 180 (d) ₹ 175

[SSC CGL Tier-II CBE, 2016]

54. Christy donated 10% of his income to an orphanage and deposited 20% of the remainder in his bank. If he has now ₹ 7200 left, what is his income.

(a) ₹ 10000 (b) ₹ 8000
(c) ₹ 9000 (d) ₹ 8500

[SSC CPO, 2016]

55. The average salary of employees in a firm was ₹ 5200 and that of females was ₹ 4200. The mean salary of all the employees was ₹ 5000. What is the percentage of female employees?

(a) 80% (b) 20%
(c) 40% (d) 30%

[SSC CGL Tier-I (CBE), 2016]

56. If A's salary is 40% less than that of B, then how much percent is B's salary more than that of A?

(a) $33\frac{1}{3}$ (b) $66\frac{2}{3}$

(c) $33\frac{2}{3}$ (d) $66\frac{1}{3}$

[SSC CGL Tier-I (CBE), 2016]

57. A man spends 15% of his income. If his expenditure is ₹ 75, his income (in rupees) is:

(a) 400 (b) 300

(c) 750 (d) 500

[SSC CGL Tier-I (CBE), 2016]

58. If A's salary is 30% more than that of B then boy how much percent is B's salary less than that of A?

(a) 13.01% (b) 13.07%

(c) 23.07% (d) 23.01%

[SSC CGL Tier-I (CBE), 2016]

59. There are 1400 students in the school 25% of them wear spectacles and $\frac{2}{7}$ of them wear the spectacles are boys. How many girls in the school do wear the spectacles?

(a) 250 (b) 100

(c) 200 (d) 300

[SSC CGL Tier-I (CBE), 2016]

60. If 60% of the students in a school are boys and the number of girls is 812, how many boys are there in the school?

(a) 1128 (b) 1218

(c) 1821 (d) 1281

[SSC CGL Tier-I (CBE), 2016]

61. The average marks obtained in a class of 50 students is 70%. The average of first 25 is 60%. and that of 24 is 80%. What is the marks obtained by the last student?

(a) 90% (b) 60%

(c) 80% (d) 70%

[SSC CGL Tier-I (CBE), 2016]

62. In an examination 65% of students pass in History and 55% students pass in Hindi. If 5% students fall in both the subjects, then what is the percentage of students who have passed in both the subjects?

(a) 15 (b) 20

(c) 25 (d) 30

[SSC CGL Tier-I CBE, 2016]

63. Due to a price hike of 20% 4 kg less sugar is available for ₹ 120. What is the initial price ₹ 120. What is the initial price per kg of sugar?

(a) ₹ 5 per kg
(b) ₹ 4 per kg
(c) ₹ 6 per kg
(d) ₹ 5.5 per kg

[SSC CGL Tier-I (CBE), 2016]

64. The population of a town is 9000. It the number of females increase by 5% and the males by 7.5% what will be the total population after. The number of females currently is 3000.

(a) 9600 (b) 9200

(c) 10500 (d) 9540

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2016]

65. The population of a city is 2000. It increased by 20% during the first year and 30% during the second year. The population after two years will be:

(a) 32000 (b) 40000

(c) 31200 (d) 30000

[SSC CGL Tier-I (CBE), 2016]

66. A set A consists of integers 27, 28, 30, 32 and 33. If integers k is included in the average of set A will increase by 30%. What is the value of integer K?

(a) 68 (b) 79

(c) 84 (d) 92

[SSC CPO SI & ASI, 2016]

67. An army lost 10% of its men in war, 10% of the remaining died due to disease and 10% of the rest were declared disabled and 10% of the rest were declared disabled. Thus the strength of the army was reduced to 7,29,000 active men. The original strength of the army was

(a) 1500000 (b) 1000000

(c) 1200000 (d) 1100000

[SSC CGL Tier-II CBE, 2016]

68. Starting with 8000 workers, the company Increase the number of workers by 5%, 10% and 20% at the end of first, second and third year respectively. The number of workers in the fourth year was

(a) 10188 (b) 11088

(c) 11008 (d) 11808

[SSC CGL Tier-I (CBE), 2016]

69. If "basis points" are defined so that 1 per cent is equal to 100 basis points, then by how many basis points is 82.5 per cent greater than 62.5 per cent?

(a) 0.2 (b) 20
(c) 200 (d) 2000

[SSC CGL Tier-I (CBE), 2016]

70. In the last financial year, a car company sold 41,800 cars. In this year, the target is to sell 51,300 cars, By what per cent must the sale be increased?

(a) $11\frac{9}{22}\%$ (b) $8\frac{9}{22}\%$
(c) $8\frac{11}{23}$ (d) $22\frac{8}{11}$

[SSC CGL Tier-I (CBE), 2016]

71. In a motor of 120 machines parts, 5% parts were defective. In another motor of 80 machine parts, 10% parts were defective. For the two motors considered together, the percentage of defective machine parts was

(a) 7 (b) 6.5
(c) 7.5 (d) 8

[SSC CGL Tier-I (CBE), 2016]

72. Number of appeared candidates from State Q increased by 100% from 2006 to 2007. If the total number of qualified candidates from State Q in 2006 and 2007 together is 408, what is the number of appeared candidates from State Q in 2006?

(a) 380 (b) 360
(c) 340 (d) 320
(e) 300

[IBPS, 2015]

73. If 60% of A = 30% of B, B = 40% of C and C = x% of A, then value of x is

(a) 800 (b) 500
(c) 300 (d) 200

[SSC, 2015]

74. Water tax is increased by 20% but its consumption is decreased by 20%. Then the increase of decrease in the expenditure of the money is

(a) 4% decrease (b) 4% increase
(c) 5% decrease (d) No change

[SSC, 2015]

75. In an office, 40% of the staff is female, 70% of the female staff and 50% of the male staff are married. The percentage of the unmarried staff in the office is

(a) 42 (b) 64
(c) 60 (d) 54

[SSC, 2015]

76. A sum of ₹7,930 is divided into 3 parts and given on loan at 5% simple interest to A, B and C for 2, 3 and 4 years respectively. If the amounts of all three are equal after their respective periods of loan, then the A received a loan of

(a) ₹2,760 (b) ₹2,750
(c) ₹2,800 (d) ₹3,050

[SSC, 2015]

77. The percentage increase in the surface area of a cube when each side is doubled is:

(a) 300% (b) 150%
(c) 50% (d) 200%

[SSC, 2015]

78. A man spends 75% of his income. His income is increased by 20% and he increased his expenditure by 10%. His savings are increased by:

(a) 25% (b) $37\frac{1}{2}$
(c) 50% (d) 10%

[SSC, 2015]

79. The percentage of metals in a mine of lead ore is 60%. Now the percentage of silver is 3/4 % of metals and the rest is lead. If the mass of ore extracted from this mine is 8000kg, the mass (in kg.) of lead of:

(a) 4762 (b) 4763
(c) 4764 (d) 4761

[SSC, 2015]

80. In an examination, a student must get 36% marks to pass. A student who gets 190 margks failed by 35 marks. The total marks in that examination is:

(a) 500 (b) 625
(c) 810 (d) 450

[SSC, 2015]

81. The number that is to be added to 10% of 320 to have the sum as 30% of 230 is:

(a) 37 (b) 32
(c) 23 (d) 73

[SSC, 2014]

82. The strength of a school increases and decreases in every alternate year by 10%. It started with increase in 2000. Then the strength of the school in 2003 as compared to that in 2000 was:

(a) Increased by 8.9%
(b) Decreased by 8.9%
(c) Increased by 9.8%
(d) Decreased by 9.8%

[SSC, 2014]

83. Two years ago, the value of a motorbike was ₹62,500. If the value depreciates by 4% every year, now its value is:

(a) ₹56,700 (b) ₹57,600
(c) ₹57,500 (d) ₹55,700

[SSC, 2014]

84. to attract more visitors, Zoo authority announces 20% discount on every ticket which cost 25 p. for this reason, sale of ticket increases by 28%. Find the percentage of increase in the number of visitors.

(a) 40% (b) 50%
(c) 60% (d) No change

[SSC, 2014]

85. In two successive years, 80 and 60 students of a school appeared at the final examination of which 60% and 80% passed respectively. The average rate of students passed (in percent) is

(a) 68% (b) $68\frac{4}{7}\%$

(c) 70% (d) $72\frac{3}{7}\%$

[SSC, 2014]

86. If A's salary is 50% more than that of B, then B's salary is less than A's by

(a) 33 % (b) $40\frac{1}{3}\%$

(c) $45\frac{1}{3}\%$ (d) $33\frac{1}{3}\%$

[SSC, 2014]

87. Two numbers A and B are such that the sum of 5% of A and 4% of B is $\frac{2}{3}$rd of the sum of 6 % of A and 8 % of B. The ratio A : B is

(a) 4 : 3 (b) 3 : 4
(c) 1 : 1 (d) 2 : 3

[SSC, 2014]

88. A man spends 75% of his income. His income increases by 20% and his expenditure also increases by 10%. The percentage of increase in his savings is

(a) 40 % (b) 30 %
(c) 50 % (d) 25 %

[SSC, 2014]

89. 20% raise of price followed by a discount of 25% of the raised portion will

(a) increase the price by 15%
(b) decrease the price by 10%
(c) increase the price by 10%
(d) decrease the price by 5%

[SSC, 2014]

90. In an election, a candidate who gets 84% of the votes is elected by a majority of 476 votes. What is total number of votes polled?

(a) 600 (b) 700
(c) 900 (d) 810

[SSC, 2014]

91. A number increased by $22\frac{1}{2}\%$ gives 98. The number is:

(a) 45 (b) 18
(c) 80 (d) 81

[SSC, 2013]

92. In an examination A got 25% marks more than B, B got 10% less than C and C got 25% more than D. If D got 320 marks out of 500, the marks obtained by A were:

(a) 405 (b) 450
(c) 360 (d) 400

[SSC, 2013]

93. Three sets of 40, 50 and 60 students appeared for an examination and the pass percentage was 100, 90 and 80, respectively. The pass percentage of the whole set is:

(a) $88\frac{2}{3}$ (b) $84\frac{2}{3}$

(c) $88\frac{1}{3}$ (d) $84\frac{1}{3}$

[SSC, 2013]

94. A clerk received annual salary of ₹3,660 in the year 1975. This was 20% more than his salary in 1974. What was his salary in 1974?

(a) ₹3,005 (b) ₹3,000
(c) ₹3,500 (d) ₹3,050

[SSC Assistant Grade III, 2013]

95. Out of his total income, Mr. Kapur spends 20% on house rent and 70% of the rest on household expenses. If he saves ₹1,800, what is his total income?

(a) 7,800 (b) 7,000
(c) 8,000 (d) 7,500

[SSC Assistant Grade III, 2013]

96. 134% of 3894 + 38.94 of 134 = ?

(a) 11452 (b) 10000
(c) 10452 (d) 1100
(e) None of these

[IBPS PO/MT, 2013]

97. Rama's expenditure and savings are in the ratio3:2. His income increases by 10 per cent. His expenditure also increases by 12%. His savingsincreases by:

(a) 7% (b) 10%
(c) 9% (d) 13%

[SSC Assistant Grade III, 2012]

98. Two numbers are 30% and 40% more than the third number respectively. The first number is x% of the second. Then x = ?

(a) $105\frac{2}{13}$ (b) 140

(c) $105\frac{5}{7}$ (d) $92\frac{6}{7}$

[SSC Assistant Grade III, 2012]

99. The price of cooking oil has increased by 25%. The percentage of reduction that a family should effect in the use of cooking oil, so as not to increase the expenditure on this account, is:

(a) 15% (b) 20%
(c) 25% (d) 30%

[SSC Assistant Grade III, 2012]

100. In an examination, 52% of the candidates failed in English and 42% failed in Mathematics. If 17% failed in both the subjects, then the percentage of candidates, who passed in both the subjects, was:

(a) 23 (b) 21
(c) 25 (d) 22

[SSC, 2012]

101. In an election there were only two candidates. One of the candidates secured 40% of votes and is defeated by the other candidate by 298 votes. The total number of votes polled is:

(a) 745 (b) 1460
(c) 1490 (d) 1500

[SSC, 2012]

102. 23% of 6783 + 57% of 8431 = ?

(a) 6460 (b) 6420
(c) 6320 (d) 6630
(e) 6360

[IBPS PO/MT, 2012]

103. The sum of three consecutive numbers is 2262. What is 41% of the highest number?

(a) 301.51 (b) 303.14
(c) 308.73 (d) 306.35
(e) 309.55

[IBPS PO/MT, 2012]

104. Akash scored 73 marks in subject A. He scored 56% marks in subject B and X marks in subject C. Maximum marks in each subject were 150. The overall percentage marks obtained by Akash in all three subjects together was 54%. How many marks did he score in subject C?

(a) 84 (b) 86
(c) 79 (d) 73
(e) None of these

[IBPS PO/MT, 2012]

105. When the price of sugar decreases by 10%, a man could buy 1 Kg more for ₹270. Then the original price of sugar per Kg is:

(a) ₹25 (b) ₹30
(c) ₹27 (d) ₹32

[SSC (GL), 2011]

106. First and second numbers are less than a third number by 30% and 37%, respectively. The second number is less than the first by:

(a) 7% (b) 4%
(c) 3% (d) 10%

[SSC (GL), 2011]

107. The price of a commodity rises from ₹6 per Kg to ₹7.50 per Kg. If the expenditure cannot increase, the percentage of reduction in consumption:

(a) 15 (b) 20
(c) 25 (d) 30

[SSC (GL), 2011]

108. Raman scored 456 marks in an examination and Seeta got 54% marks in the same examination which is 24 marks less than Raman. If the minimum passing marks in the examination is 34%, then how much more marks did Raman score than the minimum passing makrs?

(a) 184 (b) 196
(c) 190 (d) 180

[Bank of Baroda PO Examination, 2011]

109. In an Entrance Examination Ritu scored 56% marks, Smita scored 92% marks and Rina scored 634 marks. The maximum marks of the examination are 875. What are the average marks scored by all the three girls together?

(a) 929 (b) 815
(c) 690 (d) 643

[IBPS Bank PO, 2011]

110. In a test, a candidate secured 468 marks out of maximum marks '*A*'. If the maximum marks '*A*' were converted to 700 marks, he would have secured 336 marks. What were the maximum marks of the test?

(a) 775 (b) 875
(c) 975 (d) 1075

[IBPS Bank PO, 2011]

111. An HR Company employs 4800 people, out of which 45% are males and 60% of the males are either 25 years or older. How many males are employed in the company who are younger than 25 years?

(a) 2640 (b) 2160
(c) 1296 (d) 864

[IBPS Bank PO, 2011]

112. Six-elevenths of a number is equal to 22% of second number. Second number is equal to the one-fourth of third number. The value of the third number is 2400. What is the 45% of first number?

(a) 109.8 (b) 111.7
(c) 117.6 (d) None of these

[IBPS Bank PO, 2011]

113. A jar contains 10 red marbles and 30 green ones. How many red marbles must be added to the jar so that 60% of the marbles will be red?

(a) 25 (b) 30
(c) 35 (d) 40

[SSC, 2011]

114. If a number multiplied by 25% of itself gives a number which is 200% more than number, then the number is:

(a) 12 (b) 16
(c) 20 (d) 24

[SSC, 2011]

115. The price of onions has been increased by 50%. In order to keep the expenditure on onions the same the percentage of reduction in consumption has to be:

(a) 50% (b) $33\frac{1}{3}\%$
(c) 33% (d) 30%

[SSC, 2011]

Directions (Q. 116): What approximate value should come in place of question mark (?) in the following question? (Note: You are not expected to calculate the exact value.)

116. 39.897% of 4331 + 58.779% of 5003 =?

(a) 4300 (b) 4500
(c) 4700 (d) 4900
(e) 5100

[IBPS PO/MT, 2011]

117. Ramola's monthly income is three times Ravina's monthly income. Ravina's monthly income is fifteen percent more than Ruchira's monthly income. Ruchira's monthly income is ₹32,000. What is Ramola's annual income?

(a) ₹1,10.400 (b) ₹13,24,800
(c) ₹36,800 (d) ₹52,200
(e) None of these

[IBPS PO/MT, 2011]

118. In a test, a candidates secured 468 marks out of maximum marks 'A'. Had the maximum marks 'A' converted to 700, he would have secured 336 marks. What was the maximum marks of the test?

(a) 775 (b) 875
(c) 975 (d) 1075
(e) None of these

[IBPS PO/MT, 2011]

119. Six-elevenths of a number is equal to 22 per cent of the second number. The second number is equal to one-fourth of the third number. The value of the third number is 2400. What is 45% of the first number?

(a) 109.8 (b) 111.7
(c) 117.6 (d) 123.4
(e) None of these

[IBPS PO/MT, 2011]

Directions (Q. 120): What will come in place of question mark (?) in the following question?

120. 32.05% of 259.99 =?

(a) 92 (b) 88
(c) 78 (d) 90
(e) 83

[SBI Associates Banks PO, 2011]

121. Mr X invested a certain amount in Debt and Equity Funds in the ratio of 4:5. At the end of one year, he earned a total dividend of 30% on his investment. After one year, he reinvested the amount including the dividend in the ratio of 6:7 in Debt and Equity Funds. If the amount reinvested in Equity Funds was ₹94,500,

what was the original amount invested in Equity Funds?

(a) ₹75,000 (b) ₹81,000
(c) ₹60,000 (d) ₹65,000
(e) None of these

[SBI Associates Banks PO, 2011]

122. The product of one-third of a number and 150% of another number is what per cent of the product of the original numbers?

(a) 80% (b) 50%
(c) 75% (d) 120%
(e) None of these

[SBI Associates Banks PO, 2011]

123. Mr Shamin's salary increases every year by 10% in June. If there is no other increase or reduction in the salary and his salary in June 2011 was ₹22,385, what was his salary in June 2009?

(a) ₹18,650 (b) ₹18.000
(c) ₹19,250 (d) ₹18,500
(e) None of these

[SBI Associates Banks PO, 2011]

124. How many students passed in first class?

Statements:

I. 85% of the students who appeared in examination have passed either in first class or in second class or in pass class.

II. 750 students have passed in second class.

III. The number of students who passed in pass class is 28% of those passed in second class.

(a) All I, II and III
(b) Only I and III
(c) Only II and III
(d) Question cannot be answered even with information in all three statements.
(e) None of these

[SBI Associates Banks PO, 2011]

125. 34.5% of 1800 + 12.4% of 1500 = $(?)^3$ + 78:

(a) 27 (b) 9
(c) 81 (d) 162
(e) None of these

[Indian Overseas Bank PO, 2011]

126. 67% of 801 – 231.17 = ? –23% of 789:

(a) 490 (b) 440
(c) 540 (d) 520
(e) 590

[Indian Overseas Bank PO, 2011]

127. Five-ninths of a number is equal to twenty five per cent of the second number. The second number is equal to one-fourth of the third number. The value of the third number is 2960. What is 30 per cent of the first number?

(a) 88.8 (b) 99.9
(c) 66.6 (d) Cannot be determined
(e) None of these

[Indian Overseas Bank PO, 2011]

128. Dinesh's monthly income is four times Suresh's monthly income. Suresh's monthly income is twenty per cent more than Jyoti's monthly income. Jyoti's monthly income is ₹22,000. What is Dinesh's monthly income?

(a) ₹1,06,500 (b) ₹1,05,600
(c) ₹1,04,500 (d) ₹1,05,400
(e) None of these

[Indian Overseas Bank PO, 2011]

129. In a school there are 250 students, out of whom 12 per cent are girls. Each girl's monthly fee is ₹450 and each boy's monthly fee is 24 per cent more than that of a girl. What is the total monthly fee of girls and boys together?

(a) ₹1,36,620 (b) ₹1,36,260
(c) ₹1,32,660 (d) ₹1,32,460
(e) None of these

[Indian Overseas Bank PO, 2011]

130. A sum of ₹731 is distributed among A, B and C, suchthat A receives 25% more than B and B receives 25% less than C. What is C's share in the amount?

(a) ₹172 (b) ₹200
(c) ₹262 (d) ₹258
(e) None of these

[Andhra Bank PO, 2011]

131. Pradeep invested 20% more than Mohit. Mohit invested 10% less than Raghu. If the total sum of their investment is ₹17,880, how much amount did Raghu invest?

(a) ₹6,000 (b) ₹8,000
(c) ₹7,000 (d) ₹5,000
(e) None of these

[Corporation Bank PO, 2011]

132. If the numerator of a fraction is increased by 150% and the denominator of the fraction is increase by 300%, the resultant fraction is $\frac{5}{18}$. What is the original fraction?

(a) $\frac{4}{9}$ (b) $\frac{4}{5}$

(c) $\frac{8}{9}$ (d) $\frac{8}{11}$

(e) None of these

[Punjab and Sind Bank PO, 2011]

133. The price of an article was first increased by 10% and then again by 20%. If the last increased price be ₹33, the original price was:

(a) ₹30 (b) ₹27.50

(c) ₹26.50 (d) ₹25

[SSC (GL), 2010]

134. If an electricity bill is paid before due date, one gets a reduction of 4% on the amount of the bill. By paying the bill before due date a person got a reduction of ₹13. The amount of his electricity bill was:

(a) ₹125 (b) ₹225

(c) ₹325 (d) ₹425

[SSC (GL), 2010]

135. In a test, minimum passing percentage for girls and boys is 35% and 40% respectively. A boy scored 483 marks and failed by 117 marks. What are the minimum passing marks for girls?

(a) 425 (b) 520

(c) 500 (d) None of these

[CBI (PO), 2010]

136. If the numerator of a fraction is increased by 150% and the denominator of the fraction is increased by 350%, the resultant fraction is $\frac{25}{51}$, what is the original fraction?

(a) $\frac{11}{17}$ (b) $\frac{11}{15}$

(c) $\frac{15}{17}$ (d) $\frac{13}{15}$

(e) None of these

[Corporation Bank PO, 2010]

137. When the price of a toy was increased by 20%, the number of toys sold was decreased by 15%. What was its effect on the total sales of the shop?

(a) 2% increase (b) 2% decrease

(c) 4% increase (d) 4% decrease

[SSC, 2010]

138. Krishnamurthy earns ₹15000 per month and spends 80% of it. Due to pay revision, his monthly income has increased by 20% but due to price rise, he has to spend 20% more. His new savings are:

(a) ₹3,400 (b) ₹3,000

(c) ₹3,600 (d) ₹4,000

[SSC, 2010]

139. Two numbers are respectively $12\frac{1}{2}$ % and 25%more than a third number. The first number is how much per cent of the second number?

(a) 90 (b) 87.5

(c) 25 (d) 12.5

[SSC, 2010]

140. Population of a town increases 2.5% annually but is decreased by 0.5% every year due to migration. What will be the percentage of increase in 2 years?

(a) 5 (b) 4.04

(c) 4 (d) 3.96

[SSC, 2010]

141. A merchant has announced 25% rebate on prices of ready-made garments at the time of sale. If a purchaser needs to have a rebate of ₹400, then how many shirts, each costing ₹320 should he purchase?

(a) 10 (b) 7

(c) 6 (d) 5

[SSC, 2010]

142. A reduction of 10% in the price of tea enables a dealer to purchase 25 Kg more tea for ₹22500. What is the reduced price per Kg of tea?

(a) ₹70 (b) ₹80

(c) ₹90 (d) ₹100

[SSC, 2010]

143. Ram donated 4% of his income to a charity and deposited 10% of the rest in a Bank. If now he has ₹8640 left with him, then his income is:

(a) ₹12,500 (b) ₹12,000

(c) ₹10,500 (d) ₹10,000

[SSC, 2010]

144. Twelve per cent of Kaushal's monthly salary is equal to sixteen per cent of Nandini's monthly salary. Suresh's monthly salary is half that of Nandini's monthly salary. If Suresh's annual salary is ₹1.08 Lakhs, what is Kaushal's monthly salary?

(a) ₹20,000 (b) ₹18,000

(c) ₹26,000 (d) ₹24,000

(e) None of these

[Central Bank of India PO, 2010]

145. Rita invested 25% more than Sunil. Sunil invested 30% less than Abhinav, who invested ₹6,000. What is the ratio of the amount that Rita invested to the total amount invested by all of them together?

(a) 35:104 (b) 13:29
(c) 101:36 (d) 35:103
(e) None of these

[Punjab and Sind Bank PO, 2010]

146. 15% of 578 + 22.5% of 644 = ?
(a) 213.4 (b) 233.6
(c) 231.8 (d) 231.6
(e) None of these

[Indian Bank PO, 2010]

147. Sonu invested 10% more than Mona. Mona invested 10% less than Raghu. If the total sum of their investments is ₹5,780, how much amount did Raghu invest?
(a) ₹2,010 (b) ₹2,000
(c) ₹2,100 (d) ₹2,210
(e) None of these

[Indian Bank PO, 2010]

ANSWER KEYS

EXERCISE-1

1. (a)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (c)	**11.** (c)	**12.** (b)	**13.** (c)
14. (a)	**15.** (b)	**16.** (b)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (c)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (a)	**26.** (b)
27. (c)	**28.** (b)	**29.** (b)	**30.** (a)	**31.** (b)	**32.** (a)	**33.** (b)	**34.** (a)	**35.** (b)	**36.** (c)	**37.** (d)	**38.** (b)	**39.** (b)
40. (b)	**41.** (a)	**42.** (c)	**43.** (a)	**44.** (a)	**45.** (b)	**46.** (b)	**47.** (a)	**48.** (a)	**49.** (c)	**50.** (a)	**51.** (a)	**52.** (b)
53. (c)	**54.** (c)	**55.** (b)	**56.** (c)	**57.** (c)	**58.** (b)	**59.** (a)	**60.** (b)	**61.** (b)	**62.** (c)	**63.** (b)	**64.** (b)	**65.** (a)
66. (b)	**67.** (a)	**68.** (a)	**69.** (c)	**70.** (a)	**71.** (a)							

EXERCISE-2

1. (d)	**2.** (c)	**3.** (a)	**4.** (b)	**5.** (b)	**6.** (b)	**7.** (a)	**8.** (d)	**9.** (d)	**10.** (a)	**11.** (d)	**12.** (c)	**13.** (a)
14. (c)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (a)	**19.** (b)	**20.** (b)	**21.** (d)	**22.** (c)	**23.** (c)	**24.** (b)	**25.** (b)	**26.** (a)
27. (b)	**28.** (c)	**29.** (b)	**30.** (c)	**31.** (d)	**32.** (c)	**33.** (a)	**34.** (d)	**35.** (c)	**36.** (a)	**37.** (b)	**38.** (c)	**39.** (a)
40. (a)	**41.** (b)	**42.** (c)	**43.** (b)	**44.** (c)	**45.** (c)	**46.** (a)	**47.** (b)	**48.** (c)	**49.** (a)	**50.** (a)	**51.** (b)	**52.** (b)
53. (c)	**54.** (a)	**55.** (b)	**56.** (b)	**57.** (d)	**58.** (c)	**59.** (a)	**60.** (b)	**61.** (c)	**62.** (c)	**63.** (a)	**64.** (a)	**65.** (c)
66. (c)	**67.** (b)	**68.** (b)	**69.** (d)	**70.** (d)	**71.** (a)	**72.** (c)	**73.** (b)	**74.** (a)	**75.** (a)	**76.** (a)	**77.** (d)	**78.** (b)
79. (c)	**80.** (b)	**81.** (a)	**82.** (a)	**83.** (b)	**84.** (c)	**85.** (b)	**86.** (d)	**87.** (a)	**88.** (c)	**89.** (b)	**90.** (b)	**91.** (c)
92. (b)	**93.** (a)	**94.** (d)	**95.** (d)	**96.** (c)	**97.** (a)	**98.** (d)	**99.** (b)	**100.** (a)	**101.** (c)	**102.** (e)	**103.** (e)	**104.** (b)
105. (b)	**106.** (d)	**107.** (b)	**108.** (a)	**109.** (d)	**110.** (c)	**111.** (d)	**112.** (d)	**113.** (c)	**114.** (a)	**115.** (b)	**116.** (c)	**117.** (b)
118. (c)	**119.** (e)	**120.** (e)	**121.** (a)	**122.** (b)	**123.** (d)	**124.** (d)	**125.** (b)	**126.** (a)	**127.** (b)	**128.** (b)	**129.** (b)	**130.** (e)
131. (a)	**132.** (a)	**133.** (d)	**134.** (c)	**135.** (d)	**136.** (c)	**137.** (a)	**138.** (c)	**139.** (a)	**140.** (c)	**141.** (d)	**142.** (c)	**143.** (d)
144. (d)	**145.** (d)	**146.** (d)	**147.** (b)									

EXPLANATORY ANSWERS

EXERCISE-I

1. (a) $5\frac{1}{4} = \frac{21}{4} = \frac{21}{4} \times 100 = 525\%$

2. (b) $0.005 = \frac{5}{1000} = \frac{5}{1000} \times 100 = \frac{1}{2}\%$

3. (b) $6\frac{2}{3}\% = \frac{\left(\frac{20}{3}\right)}{100} = \left(\frac{20}{3} \times \frac{1}{100}\right) = \frac{1}{15}$.

4. (c) $0{\cdot}6\% = \frac{0{\cdot}6}{100} = \frac{6}{1000} = \frac{3}{500}$.

5. (b) $.025 = \left(\frac{25}{1000} \times 100\right)\% = 2.5\%$

6. (c) Let x% of 12 = 84

$\Rightarrow \frac{x}{100} \times 12 = 84$

$\Rightarrow x = \frac{84 \times 100}{12} = 700.$

$\therefore$ 700% of 12 is 84.

7. (b) $\frac{7}{8} = \left(\frac{7}{8} \times 100\right)\% = \frac{175}{2}\% = 87\frac{1}{2}\%$

8. (a) $8\frac{1}{3}\% = \frac{25}{3}\% = \frac{25}{3} \times \frac{1}{100} = \frac{1}{12}$.

9. (c) $37\frac{1}{2}\%$ of ₹48 $= 48 \times \frac{75}{2 \times 100} =$ ₹18.

Alternative Solution

$37\frac{1}{2}\%$ of ₹48

$37\frac{1}{2}\% = \frac{3}{8}$

$\therefore \frac{3}{8} \times 48 =$ ₹18

10. (c) Let x% of $\frac{2}{7} = \frac{1}{35}$

$\Rightarrow x = \frac{100 \times 7}{2 \times 35} = 10.$

$\therefore$ 10% of $\frac{2}{7}$ is $\frac{1}{35}$.

11. (c) Let 75% of 480 = $x \times 15$.

Then, $\frac{75}{100} \times 480 = 15x$

or, $x = \frac{75 \times 480}{100 \times 15} = 24.$

12. (b) Let 200% of $x = 90 \Rightarrow \frac{200}{100} \times x = 90$

$\Rightarrow x = \frac{100 \times 90}{200} = 45.$

$\therefore$ 80% of 45 $= \frac{80}{100} = 45 = 36.$

13. (c) Let the number be x, then

$37\frac{1}{2}\%$ of $x = 45 \Rightarrow \frac{75}{2} \times \frac{1}{100} \times x = 45$

or, $\frac{3}{8}x = 45 \Rightarrow x = \frac{45 \times 8}{3} = 120.$

$\therefore 87\frac{1}{2}\%$ of 120 $= \frac{75}{2} \times \frac{1}{100} \times 120 = 105.$

14. (a) Let $x \times 15 = 37.5\%$ of 220

$\Rightarrow 15x = \frac{37.5}{100} \times 220$

$\Rightarrow x = \frac{37.5 \times 220}{15 \times 100} = 5.5.$

15. (b) Let x% of 4 Km = 8 metre

$\Rightarrow \frac{x}{100} \times 4000 = 8$ ($\because$ 1 Km = 1000 metre)

$\Rightarrow x = \frac{8 \times 100}{4000} = \frac{1}{5} = 0.2$

$\therefore$ 0.2% of 4 Km = 8 metre.

16. (b) x% of $y + y$% of $x = \left(\frac{x}{100} \times y\right) + \left(\frac{y}{100} \times x\right)$

$= \frac{2}{100}xy = 2\%$ of xy.

17. (c) $0.35\% = 0{\cdot}35 \times \frac{1}{100} = 0.0035.$

18. (b) We have, 8% of x = 4% of y

$\Rightarrow \frac{8}{100} \times x = \frac{4}{100} \times y \Rightarrow x = \left(\frac{4}{100} \times \frac{100}{8}\right)y = \frac{y}{2}$.

$\therefore$ 20% of $x = \frac{20}{100} \times x = \frac{20}{100} \times \frac{y}{2}$

$= \frac{10}{100} \times y$

$= 10\%$ of y.

19. (b) Let the missing figure be a.

We have, $\frac{x}{100} \times y + \frac{a}{100} \times x = \frac{x}{100} \times (x + y)$

$\Rightarrow xy + ax = x(x + y)$

$\Rightarrow ax = x^2$. $\therefore \quad a = x$.

20. (c) We have, $x = 125\%$ of $y \Rightarrow x = \frac{125}{100}y = \frac{5}{4}y$

or, $y = \frac{4}{5}x = 0.8x$.

21. (b) 25% of 25% $= \frac{25}{100} \times \frac{25}{100} = \frac{625}{10000} = 0.0625$.

22. (c) Required number = 80 – 60% of 80

$= 80 - \frac{60}{100} \times 80 = 32$.

23. (b) 20% of 30% of 20% of 850

$= \frac{20}{100} \times \frac{30}{100} \times \frac{20}{100} \times 850 = \frac{1020}{100} = ₹10.20$.

24. (c) $16\frac{2}{3}\% = \frac{50}{3} \times \frac{1}{100} \times \frac{1}{6} = 0.167 < 0.3$.

$6\frac{2}{3}\% = \frac{20}{3} \times \frac{1}{100} = \frac{1}{15} = 0.067 < 0.3$.

$\therefore$ 0.3 is greatest.

25. (a) 40% of 20% $= \frac{40}{100} \times \frac{20}{100} = \frac{8}{100} = 8\%$

30% of 25% $= \frac{30}{100} \times \frac{25}{100} = \frac{75}{100} = 7.5\%$

and, 50% of 28% $= \frac{50}{100} \times \frac{28}{100} = \frac{14}{100} = 14\%$

$\therefore$ (40% of 20% + 30% of 25% + 50% of 28%)

$= 8\% + 7.5\% + 14\% = 29.5\%$.

26. (b) We have, $\frac{90}{100} \times A = \frac{30}{100} \times B$

$= \frac{30}{100} \times \frac{x}{100} \times A$

$\therefore \quad x = \left(100 \times \frac{100}{30} \times \frac{90}{100}\right) = 300$.

Alternative Solution

$90\%\ A = 30\%\ B$

$\frac{A}{B} = \frac{1}{3}$

$B = 3A$

$\therefore x = 300$

27. (c) Let 1 quintal 25 Kg = x% of 1 metric tonne

$\Rightarrow x\% = \frac{125}{1000} = \frac{1}{8} = \frac{1}{8} \times 100\%$

$= 12\frac{1}{2}\%$

28. (b) We have, 12% of x = 6% of y

$\Rightarrow$ 2% of x = 1% of y

$\Rightarrow (2 \times 9)\%$ of $x = (1 \times 9)\%$ of y

$\Rightarrow$ 18% of x = 9% of y.

29. (b) Here, $x = 20$.

$\therefore$ Required answer $= \left(\frac{x}{100 + x} \times 100\right)\%$

$= \left(\frac{20}{100 + 20} \times 100\right)\% = 16\frac{2}{3}\%$

30. (a) Here, $x = 25$.

$\therefore$ Required answer $= \left(\frac{x}{100 - x} \times 100\right)\%$

$= \left(\frac{25}{100 - 25} \times 100\right)\%$

$= 33\frac{1}{3}\%$

Alternative Solution

(a) Let B's income = ₹100

$\therefore$ A's income = 75% (100) = ₹75

Req. % $= \frac{25}{75} \times 100 = 33\frac{1}{3}\%$

31. (b) Here, $l = 7$ and $m = 28$.

$\therefore$ First number $= \frac{l}{m} \times 100\%$ of second number

$= \frac{7}{28} \times 100\%$ of second number

or, 25% of second number.

32. (a) Here, $x = 60$ and $y = 20$.

$\therefore$ Second number $= \left(\frac{100 + y}{100 + x} \times 100\right)\%$ of the first

$= \left(\frac{100 + 20}{100 + 60} \times 100\right)\%$ of the first

i.e., 75% of the first.

33. (b) Here, $x = 30$ and $y = 37$.

$\therefore$ Second number $= \left(\frac{100 - y}{100 - x} \times 100\right)\%$ of the first

$= \left(\frac{100 - 37}{100 - 30} \times 100\right)\%$ of the first

i.e., 90% of the first.

34. (a) Here $x = 20$ and $y = 10$.

$\therefore$ First number $= \left(\frac{100+x}{100+y}\times 100\right)\%$ of the second

$= \left(\frac{100+20}{100+10}\times 100\right)\%$ of the second

i.e., $109\frac{1}{11}\%$ of the second.

$\therefore$ The first number is $9\frac{1}{11}\%$ more than the second.

35. (b) Reduction in consumption

$= \left(\frac{P}{100+P}\times 100\right)\%$

$= \left(\frac{15}{100+15}\times 100\right)\%$ or $13\frac{1}{23}\%$

Alternative Solution

(b) Price $\uparrow = 15\% \Rightarrow \frac{15}{100} = \frac{3}{20}$

$\frac{a}{b} = \frac{3}{20}$

Consumption $\downarrow = \left(\frac{a}{a+b}\times 100\right)\%$

$= \left(\frac{3}{23}\times 100\right)\%$

$= 13\frac{1}{23}$

36. (c) Increase in consumption

$= \left(\frac{P}{100-P}\times 100\right)\%$

$= \left(\frac{10}{100-10}\times 100\right)\%$ or $11\frac{1}{9}\%$

Alternative Solution

(c) Price $\downarrow = 10\% = \frac{1}{10}$

$\frac{a}{b} = \frac{1}{10}$

Consumption $\uparrow = \left(\frac{a}{a-b}\times 100\right)\%$

$= \left(\frac{1}{9}\times 100\right)\% = 11\frac{1}{9}$

37. (d) Here, $x = 20$ and $y = -20$.

$\therefore$ The net % change in value

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(20-20-\frac{20\times 20}{100}\right)\%$ or -4%.

Since the sign is –ve, there is decrease in value by 4%

38. (b) Here, $x = -50$ and $y = 50$.

$\therefore$ The net % change in wages

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(-50+50-\frac{50\times 50}{100}\right)\%$

or, -25%.

Since the sign is –ve, he has a loss of 25%

Alternative Solution

(b) If there is $x\%$ $\uparrow$ the $x\%$ $\downarrow$ is made, then there will be always $\left(\frac{x}{10}\right)^2\%$ loss occur

$x = 50$

$\therefore$ loss $= \left(\frac{50}{10}\right)^2\% = 25\% \downarrow$

39. (b) Here, $x = -20$ and $y = -25$.

$\therefore$ The net % change in population

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(-20-25+\frac{20\times 25}{100}\right)$ or -40%

Since the sign is –ve, the population is decreased by 40% after two years.

40. (b) The equivalent discount of two successive discounts of 20% and 20%

$= \left(x+y+\frac{xy}{100}\right)\%$

$= \left(-20-20+\frac{20\times 20}{100}\right)\%$ or -36%

Given: 36% – 35% = ₹22.

$\therefore$ Amount of the bill = 22 × 100 = ₹2200.

41. (a) Here, $x = 25$ and $y = -12$.

$\therefore$ The net% change in original price

$= \left(x+y+\frac{xy}{100}\right)$

$= \left(25-12-\frac{25\times 12}{100}\right)\%$ or 10%

Since the sign is +ve, there is a profit of 10%

42. (c) The equivalent discount of two successive discounts of 20% and 10%

$= \left(x+y+\frac{xy}{100}\right)\% = \left(-20-10+\frac{20\times 10}{100}\right)\%$ or 28%